



DATA SCIENCE

Project Problem Statement Submission Form

Sec No - 8	Team No - 8	Project Title - Crop Yield Prediction
------------	-------------	---------------------------------------

1. Abstract

The Crop Yield Prediction project aims to develop a data-driven system that can estimate agricultural crop yield by analyzing important environmental, soil, and farming-related factors. Accurate crop yield prediction can help farmers and agricultural planners make better decisions regarding crop selection, resource allocation, and farm management. Traditional estimation methods may depend heavily on experience and may not fully capture the relationships between multiple influencing factors.

The proposed solution uses data science techniques to collect, preprocess, analyze, and model agricultural data. Relevant attributes such as crop type, rainfall, temperature, humidity, soil conditions, fertilizer usage, and other available parameters are examined to identify patterns associated with crop production. Exploratory data analysis and visualization are used to understand the dataset, while suitable machine learning algorithms can be trained and evaluated to predict expected crop yield.

By providing data-based predictions and meaningful visual insights, the system can support more informed agricultural planning. The project demonstrates the practical application of data preprocessing, exploratory analysis, visualization, feature selection, machine learning, and model evaluation to a real-world agricultural problem.

2. Module Mapping

Module Name	Assigned Roll Number(s)	Signature
Data Collection & Preprocessing	2420030467 K.V.S Jaswant	
Data Analysis & Visualization	2420030737 Teja D	
Machine Learning & Model Evaluation	2420030754 Bhargav	

3. Module Details

Module 1: Data Collection & Preprocessing – Roll No. 2420030467

Responsible for collecting the crop and agricultural dataset, checking data quality, handling missing or inconsistent values, encoding categorical variables, and preparing the data for analysis and machine learning.

Module 2: Data Analysis & Visualization – Roll No. 2420030737

Responsible for exploratory data analysis and visualization. This includes studying relationships between crop yield and factors such as rainfall, temperature, soil properties, crop type, and fertilizer usage using suitable charts and statistical summaries.

Module 3: Model Evaluation – Roll No. 2420030754

Evaluate the trained machine learning models for crop yield prediction using suitable performance metrics. Compare the predicted and actual crop yields, analyze the model performance, and identify the model that provides the most accurate predictions.

4. Frameworks / Platforms / Software Used

- **Python:** Used as the primary programming language for data preprocessing, analysis, visualization, and machine learning.
- **Pandas & NumPy:** Used for data manipulation, cleaning, numerical calculations, and dataset preparation.
- **Matplotlib / Seaborn:** Used to create visualizations for exploratory data analysis and to identify patterns and relationships in the dataset.
- **Scikit-learn:** Used for machine learning model development, data splitting, preprocessing, training, and evaluation.
- **Jupyter Notebook / Google Colab:** Used as the development environment for executing analysis and machine learning experiments.

Professor: Sunandha

Guide Name and Signature (remarks if any) _____